Electric car: launch the design

antocciphysics

UNIT 8 / DAY 1 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A single-loop toy calculation gives 3 V induced emf. Explain why this alone cannot predict the effects on a whole power grid; name two missing features.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: average speed = distance/elapsed time. Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Start Finish



Distance: ____ Timing method: _____

Marzano check - circle: ① ② ③

03 Read and interpret the model

Sketch your proposed car. Label the battery, switch, motor, driven axle and wheels. Add a theme that will not obstruct moving parts.



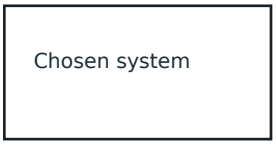
Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

04 Apply the physics

Your first design uses a 3 V supply and an expected motor current of 0.40 A. Estimate electrical input power. Label at least three destinations that prevent all input power from becoming useful forward-motion power.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Write one measurable design goal and one constraint. Explain how you will measure the goal instead of judging only by appearance.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A circuit has 4 identical series resistors. Add two more at unchanged ideal supply voltage. Compare new current with the original.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Create a fair first-test plan naming distance, timing method, number of trials and variables held fixed.

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Choose a speed target and a reliability target for a 3 m course. Design repeated trials and a rule for judging success without selecting only the fastest run.

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3